BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Random Number Generation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does WHERE do in an SQL query?

2. What does a MAC address identify?
   A) A website's domain name
   B) A network interface
   C) A user's password
   D) A file's folder

3. "A firewall confirms that every sentence on every website is true." Correct this. Explain why.

4. Data must reach another network in the right order. "TCP chooses fonts; IP draws images." Correct this. Explain each protocol's job.

Today's objectives

Use random values in programs. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

Random numbers

random(low,high) produces a value inside the inclusive range.

Integer range

random(1,6) can return any whole number from 1 to 6.

Unpredictability

Successive random calls can produce different values; outputs are not fixed.

Simulation

Random values can model dice, cards or game events without physical devices.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Random numbers

random(low,high) produces a value inside the inclusive range. In the dice game, identify where this technique applies.

Integer range

random(1,6) can return any whole number from 1 to 6. In the quiz bonus round, identify where this technique applies.

Unpredictability

Successive random calls can produce different values; outputs are not fixed. In the board-game move, identify where this technique applies.

Simulation

Random values can model dice, cards or game events without physical devices. In the random test selector, identify where this technique applies.

I Do - Worked model

A dice game uses random numbers and integer range. Explain what each contributes. [4 marks]

- Identify the program's purpose and the pieces of data needed.
- Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1].
- Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1].
- Each feature needs a correct technical fact plus an explanation.

We Do - Guided practice

1. A quiz bonus round needs random numbers. Explain the purpose. [2 marks]

2. Explain integer range for a board-game move. [2 marks]

3. When programming a random test selector, explain why unpredictability and simulation matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write one OCR ERL instruction to generate an integer from 1 to 6 inclusive.

```
roll = ...
print(roll)
```

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain integer range and unpredictability in a virtual spinner. [4 marks]

Q4. A programmer makes a sports draw. Explain the roles of random numbers, unpredictability and simulation. [6 marks]

Exit Ticket - True or False?

random(low,high) produces a value inside the inclusive range.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It selects rows that meet a condition.

2. B - A network interface

3. It uses rules to allow or block network traffic.

4. TCP supports reliable delivery in order. IP handles addressing and routing between networks. Each protocol has its own job; encryption does not prove a site is honest.

I Do - Explain the model

Identify the program's purpose and the pieces of data needed.

Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1].

Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1].

Each feature needs a correct technical fact plus an explanation.

We Do - Model answers

1. Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1].

2. Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1].

3. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1]. Simulation [1]; Random values can model dice, cards or game events without physical devices. [1].

Code First - Answer

```
roll = random(1,6)
print(roll)
```

Expected output: Any whole number from 1 to 6 inclusive (not a fixed output)

You Do - Self-mark

Q2: Random numbers [1]; Integer range [1].

Q3: Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1]. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1].

Q4: Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1]. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1]. Simulation [1]; Random values can model dice, cards or game events without physical devices. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.